

Natural Language Processing

12. Small Language Models

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Notice

Credits

The present slides are largely an adaptation of existing material, including slides by [Jurafsky & Martin](#), [Manning](#), [Potts & MacCartney](#), [Black & Mortensen](#), and others, to whom I am very grateful.

Downloading and sharing

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Reducing Model Size

Rapid Progress from Pre-Training 😊

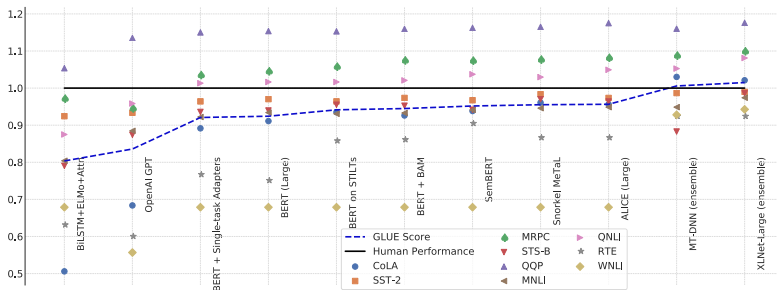
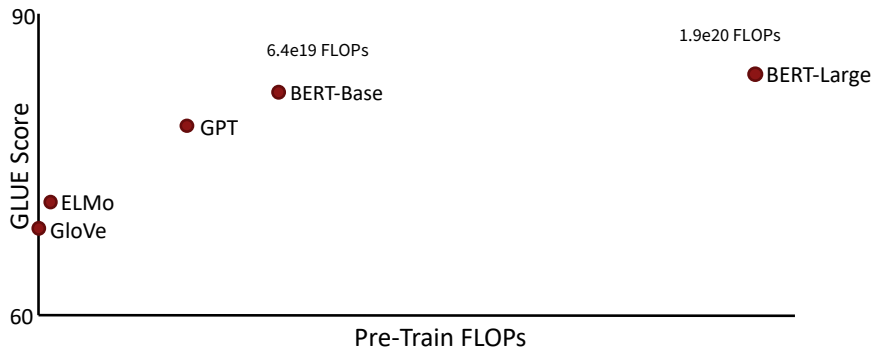


Figure 1: GLUE benchmark performance for submitted systems, rescaled to set human performance to 1.0, shown as a single number score, and broken down into the nine constituent task performances. For tasks with multiple metrics, we use an average of the metrics. More information on the tasks included in GLUE can be found in Wang et al. (2019a) and in Warstadt et al. (2018, CoLA), Socher et al. (2013, SST-2), Dolan and Brockett (2005, MRPC), Cer et al. (2017, STS-B), and Williams et al. (2018, MNLI), and Rajpurkar et al. (2016, the original data source for QNLI).

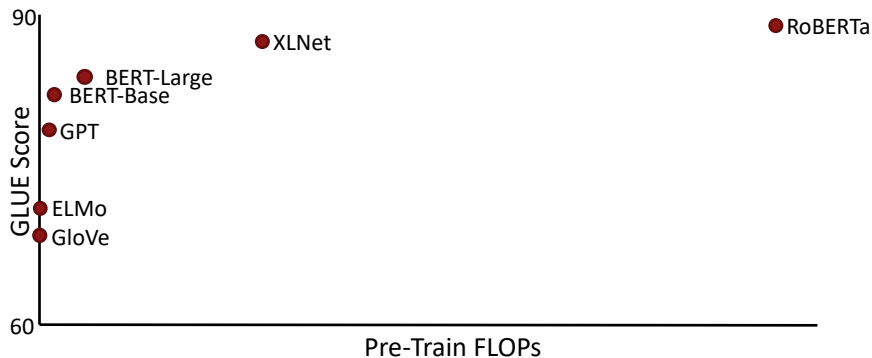
Steep Increase in Cost of Pre-Training 😞



BERT-Large uses 60x more compute than ELMo

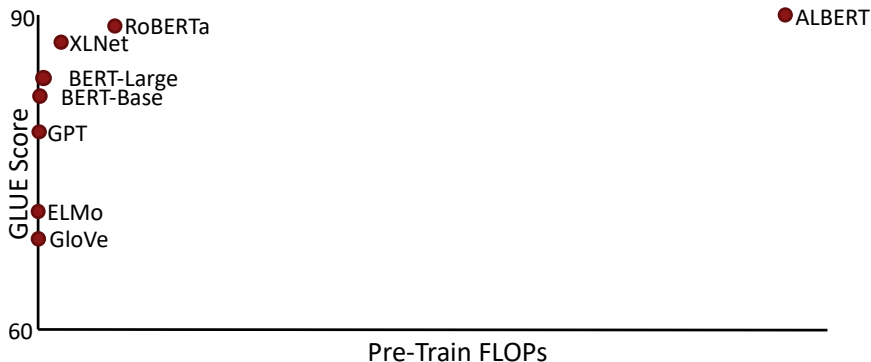
The Cost of Training NLP Models: A Concise Overview, O Sharir et al, 2020

Steep Increase in Cost of Pre-Training 😞



RoBERTa uses 16x more compute than BERT-Large

Steep Increase in Cost of Pre-Training 😞



ALBERT uses 10x more compute than RoBERTa

Steep Increase in Number of Parameters

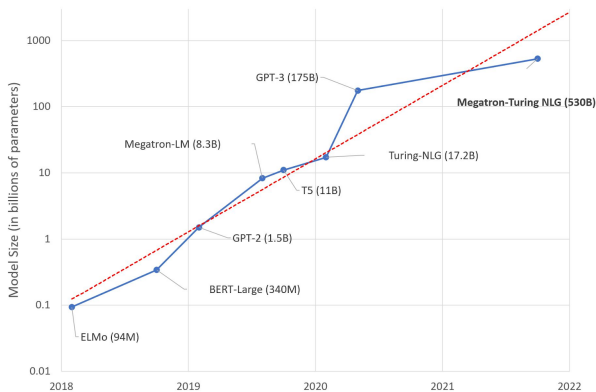


Image credits: [Huggingface](#)

- Recent large “open-weight” models have 400B+ parameters

Problems with Large Models

- Increased model size—more data, more layers, longer input sequences—generally yields better performance
- However, training requires large clusters of multi-GPU servers
- Unaffordable even for big universities and large companies
- Environmental footprint
- Even fine-tuning and maintaining LLMs for down-stream tasks is computationally expensive (latency, hardware constraints)
- **In-house solution:** low-rank adaptation, model compression
- **Service-based solution:** prompting instead of fine-tuning

The Cost of Training NLP Models: A Concise Overview, O Sharir et al, 2020

Estimating the Carbon Footprint of BLOOM, a 176B Parameter Language Model, AS Luccioni et al, 2023

The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink, D Patterson et al, 2022

DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter, V Sanh et al NeurIPS Workshops 2019

Pre-Trained Models: Past, Present and Future, X Han et al, AI Open 2021

Training Compute-Optimal Large Language Models, J Hoffmann et al, 2022

A Survey of Large Language Models, WX Zhao et al, 2023

Approaches to Reducing Model Size

Several approaches to obtain lighter and faster language models

- **Parameter sharing** between layers, projection matrices etc. (Linformer, Reformer)
- **Pruning**: removing weights with small impact on loss (*saliency*); however, sparse models unoptimized for modern HW
- **Knowledge distillation**: student trained to reproduce teacher's hidden states (DistilBERT, DistilRoBERTa, DistilBART, TinyBERT)
- **Mixed precision**: storing and computing weights, activations and gradients in single-precision or even **4-bit quantization (QLoRA)**
- **Mixture of Experts** each specialized in subset of data with router to forward input to corresponding expert (e.g.: Mixtral). Efficient for distributed systems
- Many other general/specific techniques like **sparse attention**

A Practical Survey on Faster and Lighter Transformers, Q Fournier et al, 2023

Efficient Methods for Natural Language Processing: A Survey, M Treviso et al, 2023

Knowledge Distillation

- A general compression method in DL, introduced by Hinton et al
- **Knowledge**: a learned mapping from input vectors to output vectors
- Idea: leverage relative probabilities of the **incorrect answers** to learn how teacher model generalizes
- Use **temperature** for smoother softmax distribution
- Introduce **distillation loss** = distance from soft targets

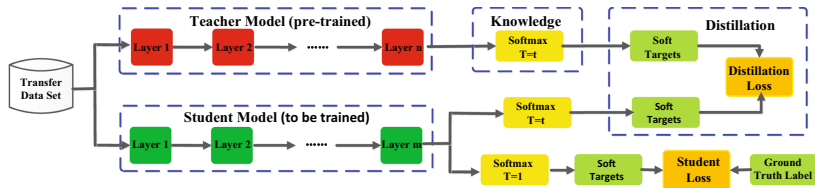


Fig. 5 The specific architecture of the benchmark knowledge distillation (Hinton et al. 2015)

Knowledge Distillation, Hinton, Vinyals, Dean, 2015

Knowledge Distillation: A Survey, Gou et al, 2021

Model Compression by Vocabulary Transfer

Pretrained LMs typically use variations of Byte-Pair Encoding tokenizers, like WordPiece, SentencePiece, BertTokenizer, etc.

- However, language shifts significantly in vertical domains
- Model compression by **ad-hoc tokenization**
- Ad-hoc tokenizers may improve model efficiency
 - by reducing **average length** of the tokenized sequences
 - by reducing size of **embedding matrix**

Summary of the tokenizers, Huggingface

Fast Vocabulary Transfer for Language Model Compression, L. Ge et al, 2022

Vocabulary Transfer

Smaller input size by ad-hoc tokenization

- **Input:** He was initially treated with interferon alfa.
- $\mathcal{T}_{gen}$: He, was, initially, treated, with, inter, ##fer, ##on, al, ##fa, .
- $\mathcal{T}_{in}$: He, was, initially, treated, with, interferon, alfa, .

Knowledge transfer using **Fast Vocabulary Transfer**

Fast Vocabulary Transfer (FVT)

Given a vertical domain $\mathcal{D}_{in}$:

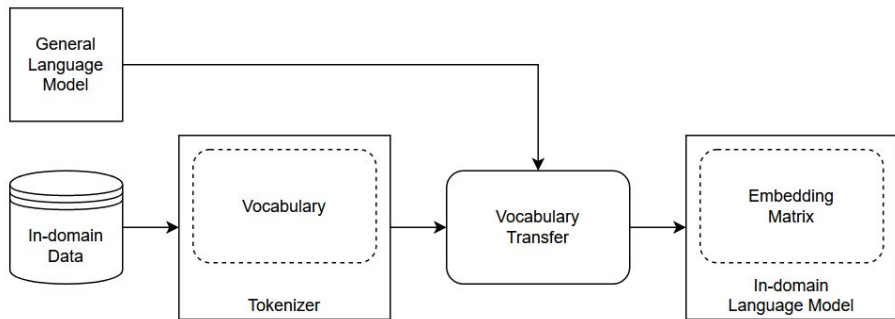
- Train a tokenizer $\mathcal{T}_{in}$ in $\mathcal{D}_{in}$
- Establish a mapping between the vocabulary V_{in} of $\mathcal{T}_{in}$ and the dictionary of the general-purpose tokenizer $\mathcal{T}_{gen}$
- Assign embeddings E_{in} transferring pre-trained representation according to the established mapping and an assignment function

FVT: Each token $t_i \in \mathcal{V}_{in}$ is partitioned using $\mathcal{T}_{gen}$, and its embedding $E_{in}(t_i)$ is assigned as follows:

$$E_{in}(t_i) = \frac{1}{|\mathcal{T}_{gen}(t_i)|} \cdot \sum_{t_j \in \mathcal{T}_{gen}(t_i)} E_{gen}(t_j)$$

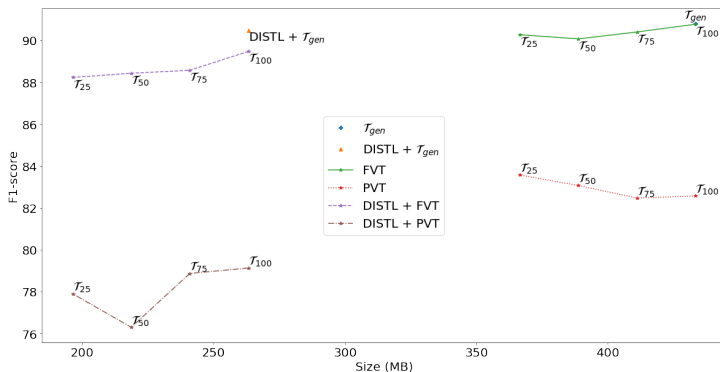
Notice: if t_i belongs to both vocabularies, then $E_{in}(t_i) = E_{gen}(t_i)$

Fast Vocabulary Transfer (FVT)



FVT + Knowledge Distillation

FVT can be combined with knowledge distillation by distilling the student first on general-purpose data and then again on in-domain data only



Distill, Adapt, Distill: Training Small, In-Domain Models for Neural Machine Translation, M Gordon et al, 2020

Questions?